

**CEBU INSTITUTE OF TECHNOLOGY
UNIVERSITY**

COLLEGE OF COMPUTER STUDIES

Software Requirements Specifications

for
Readlr

Change History

Version	Date	Description	Author
1.0	2026-05-15	Initial SRS Draft	Dwight

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1. Introduction

1.1. Purpose

The purpose of this document is to provide a detailed description of the Readlr system, a gamified, storytelling-based mobile application integrated with automated speech recognition (ASR) designed to improve phonemic awareness, pronunciation accuracy, and early reading readiness among Grade 1 learners in Philippine elementary schools. This document serves as a guide for developers, researchers, and stakeholders to ensure alignment with system requirements and educational objectives.

1.2. Scope

Readlr is a mobile-based educational application designed for Grade 1 learners that combines gamification, storytelling, and speech recognition technology to support early literacy development.

The system includes the following core functionalities:

- Story-Based Learning Adventure System
- Phonetic Challenge Module (phonemes, CVC words, sight words)
- Automated Speech Recognition (ASR) Feedback Engine
- Reward and Sticker Collection System
- Student Progress Tracking Dashboard
- Phoneme and Vocabulary Learning Bank

The application aims to function as a self-directed learning tool that enhances pronunciation and phonemic awareness through interactive gameplay.

1.3. Definitions, Acronyms and Abbreviations

- **ASR** – Automated Speech Recognition
- **MELCs** – Most Essential Learning Competencies (DepEd curriculum framework)
- **CVC** – Consonant-Vowel-Consonant words (e.g., “cat”, “dog”)
- **Phoneme** – The smallest unit of sound in speech
- **Gamification** – Use of game elements in non-game contexts
- **ISO/IEC 25010** – Software quality evaluation model
- **WCPM** – Words Correct Per Minute (reading fluency metric)
- **UI/UX** – User Interface / User Experience

1.4. References

- [1] Department of Education (DepEd), “Comprehensive Rapid Literacy Assessment,” 2025.
- [2] N. K. Librea et al., “Low reading literacy skills of elementary pupils in the Philippines,” 2023.
- [3] World Bank, “Learning Poverty Brief: Philippines,” 2022.
- [4] Y. Sun, “Impact of ASR technology on pronunciation learning,” 2023.
- [5] Southeast Asia Primary Learning Metrics (SEA-PLM), “Regional Education Report,” 2024.

[6] International Organization for Standardization, "ISO/IEC 25010:2011 Systems and software quality models," 2011.

2. Overall Description

2.1. Product perspective

Readlr is a standalone mobile application designed for early-grade literacy development. It uses a client-side application architecture integrated with speech recognition APIs to evaluate pronunciation input in real time.

The system operates offline for story progression and UI rendering, while ASR functionality may require internet connectivity depending on implementation (e.g., cloud-based speech API).

2.2. User characteristics

- **Primary Users:** Grade 1 learners (ages 6–7) with basic literacy skills and limited digital literacy experience
- **Secondary Users:** Teachers who may monitor learner progress
- **Administrators/Developers:** Responsible for system configuration and content management

Users are expected to:

- Follow visual and audio instructions
- Interact primarily through voice input
- Engage in short attention-span learning sessions

2.4. Constraints

- *Requires a device with microphone capability for speech input*
- *Dependent on ASR accuracy, which may vary due to noise and accent differences*
- *Limited to Grade 1 phonics-based vocabulary scope*
- *Requires stable internet connection if cloud-based ASR is used*
- *Must comply with child-friendly UI/UX standards*

2.5. Assumptions and dependencies

3. *Assumes availability of mobile devices in pilot school setting*
4. *Assumes learners can follow basic visual/audio instructions independently*
5. *Assumes ASR API or speech engine is available and functional during development*
6. *Depends on DepEd MELCs and Marungko-based phonics sequencing for content design*
7. *Assumes teacher support during initial testing phase*

3. Specific Requirements

7.1. External interface requirements

3.1.1. Hardware interfaces

- *Mobile devices (Android-based smartphones/tablets)*
- *Built-in microphone for speech input*
- *Speakers or headphones for audio feedback*
- *Optional internet connectivity for ASR processing*

3.1.2. Software interfaces

- *Speech Recognition API (e.g., Google Speech-to-Text or equivalent ASR service)*
- *Mobile operating system (Android minimum version TBD)*
- *Local database for storing progress and user data*
- *Game engine framework (e.g., Unity or Flutter-based system)*

3.1.3. Communications interfaces

- *HTTPS protocol for ASR requests (if cloud-based)*
- *Local device storage for offline progress tracking*
- *Optional API communication for analytics or updates*

7.2. Functional requirements

Module 1: Story-Based Learning System

1.1 Story Progression Module

- *Displays interactive narrative-based levels*
- *Guides learner through phonetic challenges*
- *Unlocks next story chapter upon completion*

Artifacts:

- *Use Case Diagram*
 - *Use Case Description*
 - *Activity Diagram*
 - *Wireframe*
-

1.2 Level Navigation Module

- *Allows learners to progress through structured phonics levels*
- *Controls difficulty progression (vowels → blends → CVC words)*

Artifacts:

- *Use Case Diagram*
- *Use Case Description*
- *Activity Diagram*
- *Wireframe*

Module 2: Speech Recognition Learning System

2.1 Pronunciation Challenge Module

- *Prompts learner to pronounce phonemes or words*
- *Captures voice input via microphone*
- *Sends input to ASR system for evaluation*

Artifacts:

- *Use Case Diagram*
- *Use Case Description*
- *Activity Diagram*

- Wireframe
-

2.2 ASR Feedback Engine

- Evaluates pronunciation using confidence score thresholds
- Triggers success animation if score is above threshold
- Activates corrective feedback loop if below threshold
- Provides:
 - correct audio model
 - visual mouth-shape guidance
 - retry prompts

Artifacts:

- Use Case Diagram
- Use Case Description
- Activity Diagram
- Wireframe

Module 3: Reward and Engagement System

3.1 Sticker Reward System

- Awards digital stickers for completed challenges
- Unlocks collectibles based on performance

Artifacts:

- Use Case Diagram
 - Use Case Description
 - Activity Diagram
 - Wireframe
-

3.2 Progress Tracking Module

- Records learner performance per session
- Displays completed levels and accuracy score
- Stores historical learning progress

Artifacts:

- Use Case Diagram
- Use Case Description
- Activity Diagram

- Wireframe

3.4 Non-functional requirements

Performance

- System should respond to voice input within 2–3 seconds
- ASR processing should maintain acceptable accuracy under moderate noise conditions
- Application should support smooth animation rendering without lag on mid-range devices

Security

- Learner data must be stored securely and locally or via encrypted transmission
- No personal identifiable data beyond basic learner profile should be required
- System must comply with child data protection principles

Reliability

- Application must function offline for core gameplay features
- ASR failure should not crash the system but trigger retry feedback
- System should maintain progress even after unexpected shutdown